

The Coordinate-Based MILP Implemented in `gurobi_coordinate_solver.py`

Abstract

This document states the mixed-integer linear model constructed by `gurobi_coordinate_solver.py`. Unlike the position-indexed model in `MILP_formulation.tex`, it does not enumerate every possible placement. Each physical box receives an integer coordinate, a pallet assignment, and an orientation. Pairwise direction variables enforce three-dimensional non-overlap. Optional support constraints use the exact horizontal intersection area of two boxes. The implemented objective is solely to minimize the number of used pallets. The document first gives the default `legacy` formulation and then records the changes made by the selectable `reduced_exact` formulation.

1 Scope and source of the model

The solver reads an MCPP/PL-100 JSON instance and a JSON configuration file. The instance supplies the pallet and physical boxes. The configuration supplies the spatial grid, maximum candidate-pallet count, rotation and support modes, symmetry options, and Gurobi stopping parameters. All spatial quantities in the optimization model are integer numbers of grid cells.

The implementation currently uses box geometry and identity information, but does *not* impose weight, food-chemical, fragility, or retrieval-priority constraints. Those attributes are parsed, used in the key that recognizes identical boxes, and reproduced in solution files. This distinction is important: their presence in the input does not mean they are active physical constraints in this coordinate model.

2 Input processing and primitive data

2.1 Grid conversion

Let $g \in \mathbb{Z}_{>0}$ be `grid_mm`. It comes from the configuration, unless its value is the string `input`, in which case it comes from `meta.grid_mm` in the instance (default 50 mm if absent).

Let $(L^{\text{mm}}, W^{\text{mm}}, H^{\text{mm}})$ be the pallet dimensions read from one of the accepted JSON schemas. The model dimensions are conservatively floored:

$$(L, W, H) = \left(\left\lfloor \frac{L^{\text{mm}}}{g} \right\rfloor, \left\lfloor \frac{W^{\text{mm}}}{g} \right\rfloor, \left\lfloor \frac{H^{\text{mm}}}{g} \right\rfloor \right). \quad (1)$$

For box i , let $(\ell_i^{\text{mm}}, w_i^{\text{mm}}, h_i^{\text{mm}})$ be its dimensions after schema normalization. Its unrotated grid dimensions are rounded upward:

$$(\ell_i, w_i, h_i) = \left(\max \left\{ 1, \left\lceil \frac{\ell_i^{\text{mm}}}{g} \right\rceil \right\}, \max \left\{ 1, \left\lceil \frac{w_i^{\text{mm}}}{g} \right\rceil \right\}, \max \left\{ 1, \left\lceil \frac{h_i^{\text{mm}}}{g} \right\rceil \right\} \right). \quad (2)$$

Flooring the container and ceiling the boxes makes the discretization conservative: a grid-feasible packing does not exceed the physical pallet boundary, apart from ordinary floating-point input tolerances used in parsing.

2.2 Sets and indices

$$\mathcal{I} := \{0, \dots, n-1\}, \quad i, j \in \mathcal{I}, \quad (3)$$

$$\mathcal{P} := \{0, \dots, P^{\max} - 1\}, \quad p \in \mathcal{P}, \quad (4)$$

$$\mathcal{E} := \{(i, j) \in \mathcal{I}^2 : i < j\}, \quad (5)$$

$$\vec{\mathcal{E}} := \{(i, j) \in \mathcal{I}^2 : i \neq j\}. \quad (6)$$

Here n is the number of records in the instance’s `items` or `boxes` array. $P^{\max}$ is `max_pallets`; for batch solves it can be raised to a benchmark-provided pallet upper bound. Candidate pallets are identical and their indices have no physical meaning.

For each i , preprocessing constructs a list of distinct permitted oriented dimension triples

$$\mathcal{O}_i = \{0, \dots, m_i - 1\}, \quad (d_{io}^x, d_{io}^y, d_{io}^z) \in \mathbb{Z}_{>0}^3. \quad (7)$$

The list comes from `rotation_mode`:

none	only (ℓ_i, w_i, h_i) ;
yaw	the original orientation and a swap of the horizontal axes;
metadata	the same yaw-only rule (and thus no extra behavior beyond yaw in the current code);
six	all distinct axis permutations, except that a box with <code>upright_only=true</code> is still restricted to yaw rotations.

Repeated triples caused by equal side lengths are removed while preserving their first occurrence.

2.3 Parameters and their provenance

Symbol	Code/input source	Meaning and use
L, W, H	instance pallet; Eq. (1)	pallet extents in grid cells and big- M constants for the three axes.
$P^{\max}$	config max_pallets	number of identical candidate pallets.
d_{io}^a	item dimensions, grid, rotation mode	oriented extent of box i on axis $a \in \{x, y, z\}$.
v_i	$\ell_i w_i h_i$	conservative grid-cell volume. Rotation does not change it.
V	LWH	pallet volume in grid cells.
α	support.minimum_fraction	supported base fraction in fraction mode; default 0.75.
(r, s)	rational approxi- mation of $\alpha = r/s$	numerator and denominator, each produced with denominator at most 1000, used to avoid a fractional coefficient in the support row.
$\bar{A}_i$	$\max_{o \in \mathcal{O}_i} d_{io}^x d_{io}^y$	largest possible base area of box i .
$A^{\max}$	LW	safe upper bound in support-area linearizations.
K^{vol}	$\max\{1, \lceil \sum_i v_i / V \rceil\}$	volume lower bound on the number of pallets.

Symbol	Code/input source	Meaning and use
M^{loc}	$L(W+1)(H+1) + W(H+1) + H$	big- M for lexicographic position ordering of identical boxes.

The instance also supplies box ID, SKU, weight, volume in dm^3 , family, food flag, chemical flag, fragile flag, upright-only flag, and retrieval priority. ID and the original millimetre dimensions are reporting fields. SKU, grid dimensions, weight, family, the three handling flags, upright-only, and priority form the identical-item grouping key. Of these, only dimensions and upright-only status otherwise affect the current optimization model.

3 Default (legacy) decision variables

The legacy formulation stores a separate local geometric state for every box–pallet combination. Variables belonging to an unassigned combination are forced to zero by the boundary equations.

Variable	Index	Domain	Meaning
u_p	$p \in \mathcal{P}$	$\mathbb{B}$	1 iff candidate pallet p is used.
a_{ip}	$i \in \mathcal{I}, p \in \mathcal{P}$	$\mathbb{B}$	1 iff box i is assigned to pallet p .
r_{iop}	$i, p, o \in \mathcal{O}_i$	$\mathbb{B}$	1 iff i is on p with orientation o .
x_{ip}, y_{ip}, z_{ip}	i, p	integer, respectively $[0, L], [0, W], [0, H]$	lower-left-bottom coordinates.
$\bar{x}_{ip}, \bar{y}_{ip}, \bar{z}_{ip}$	i, p	integer, same bounds	right, front, and top coordinates.
b_{ip}	i, p	$\mathbb{Z}_{\geq 0}$	oriented base area; zero if unassigned.
λ_{ijp}^x	$(i, j) \in \overrightarrow{\mathcal{E}}, p$	$\mathbb{B}$	1 exactly when i ends at or before j starts on x .
λ_{ijp}^y	$(i, j) \in \overrightarrow{\mathcal{E}}, p$	$\mathbb{B}$	analogous “in front of” relation on y .
λ_{ijp}^z	$(i, j) \in \overrightarrow{\mathcal{E}}, p$	$\mathbb{B}$	analogous “below” relation on z .
o_{ijp}^x, o_{ijp}^y	$(i, j) \in \mathcal{E}, p$	$\mathbb{B}$	1 iff assigned boxes overlap strictly on the named horizontal axis.
o_{ijp}^{xy}	$(i, j) \in \mathcal{E}, p$	$\mathbb{B}$	logical AND $o_{ijp}^x \wedge o_{ijp}^y$.
f_{jp}	j, p	$\mathbb{B}$	1 iff assigned box j is on the floor.
c_{ijp}	$(i, j) \in \overrightarrow{\mathcal{E}}, p$	$\mathbb{B}$	1 may certify that i directly supports j .
q_{ijp}^x, q_{ijp}^y	$(i, j) \in \mathcal{E}, p$	integer	exact horizontal overlap lengths when area support is active.
A_{ijp}	$(i, j) \in \mathcal{E}, p$	continuous or integer	exact footprint-intersection area.
S_{ijp}	$(i, j) \in \overrightarrow{\mathcal{E}}, p$	continuous or integer	$A_{\min\{i,j\}, \max\{i,j\}, p}$ when $c_{ijp} = 1$, and zero otherwise.

The area auxiliaries are continuous by default in the legacy model, although their defining equations give integral values at integer coordinates. Setting `area_auxiliary_type` to `integer` declares them integral.

Define the following convenient linear expressions:

$$D_{ip}^x := \sum_{o \in \mathcal{O}_i} d_{io}^x r_{iop}, \quad D_{ip}^y := \sum_{o \in \mathcal{O}_i} d_{io}^y r_{iop}, \quad D_{ip}^z := \sum_{o \in \mathcal{O}_i} d_{io}^z r_{iop}. \quad (8)$$

4 Objective

The single implemented optimization criterion is

$$\min \sum_{p \in \mathcal{P}} u_p. \quad (9)$$

Weight, center of mass, accessibility, family separation, and similar project attributes do not appear in this objective.

5 Assignment, orientation, and containment

Each physical box is assigned once, and it has one orientation precisely on its assigned pallet:

$$\sum_{p \in \mathcal{P}} a_{ip} = 1 \quad \forall i \in \mathcal{I}, \quad (10)$$

$$\sum_{o \in \mathcal{O}_i} r_{iop} = a_{ip} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}. \quad (11)$$

Assignment activates a pallet, and activation is exact:

$$a_{ip} \leq u_p \quad \forall i, p, \quad (12)$$

$$u_p \leq \sum_{i \in \mathcal{I}} a_{ip} \quad \forall p. \quad (13)$$

The endpoint variables are defined by

$$\bar{x}_{ip} = x_{ip} + D_{ip}^x, \quad \bar{y}_{ip} = y_{ip} + D_{ip}^y, \quad \bar{z}_{ip} = z_{ip} + D_{ip}^z, \quad (14)$$

$$b_{ip} = \sum_{o \in \mathcal{O}_i} d_{io}^x d_{io}^y r_{iop}. \quad (15)$$

Containment and the zeroing of inactive local states follow from

$$\bar{x}_{ip} \leq L a_{ip}, \quad \bar{y}_{ip} \leq W a_{ip}, \quad \bar{z}_{ip} \leq H a_{ip} \quad \forall i, p. \quad (16)$$

The volume inequalities are valid but not sufficient packing conditions:

$$\sum_{i \in \mathcal{I}} v_i a_{ip} \leq V u_p \quad \forall p, \quad (17)$$

$$\sum_{p \in \mathcal{P}} u_p \geq K^{\text{vol}}. \quad (18)$$

In addition, the implementation fixes $u_p = 1$ for $p = 0, \dots, K^{\text{vol}} - 1$. This is valid because pallet labels are interchangeable and used pallets are ordered as a prefix.

6 Pairwise non-overlap

For every ordered pair and pallet, forward big- M inequalities give the meaning of the direction variables:

$$\bar{x}_{ip} \leq x_{jp} + L(1 - \lambda_{ijp}^x), \quad (19)$$

$$\bar{y}_{ip} \leq y_{jp} + W(1 - \lambda_{ijp}^y), \quad (20)$$

$$\bar{z}_{ip} \leq z_{jp} + H(1 - \lambda_{ijp}^z). \quad (21)$$

They can be 1 only for two boxes assigned to that pallet:

$$\lambda_{ijp}^a \leq a_{ip}, \quad \lambda_{ijp}^a \leq a_{jp} \quad \forall (i, j) \in \overrightarrow{\mathcal{E}}, p, a \in \{x, y, z\}. \quad (22)$$

The reverse inequalities make each binary an exact strict-separation indicator at integer coordinates. For example,

$$x_{jp} \leq \bar{x}_{ip} - 1 + L(\lambda_{ijp}^x + 2 - a_{ip} - a_{jp}), \quad (23)$$

with analogous inequalities on y and z . If both boxes are assigned and $\lambda_{ijp}^x = 0$, Eq. (23) says $x_{jp} \leq \bar{x}_{ip} - 1$, i.e., the two integer intervals overlap strictly or j lies to the left. The one-cell term is the discrete analogue of a strict inequality.

Two boxes assigned to the same pallet must be separated on at least one axis:

$$\begin{aligned} \lambda_{ijp}^x + \lambda_{jip}^x + \lambda_{ijp}^y + \lambda_{jip}^y + \lambda_{ijp}^z + \lambda_{jip}^z \\ \geq a_{ip} + a_{jp} - 1 \quad \forall (i, j) \in \mathcal{E}, p \in \mathcal{P}. \end{aligned} \quad (24)$$

The horizontal overlap indicators used by support are defined by the standard binary AND logic

$$\begin{aligned} o_{ijp}^x &\leq a_{ip}, a_{jp}, 1 - \lambda_{ijp}^x, 1 - \lambda_{jip}^x, \\ o_{ijp}^x &\geq a_{ip} + a_{jp} - 1 - \lambda_{ijp}^x - \lambda_{jip}^x, \end{aligned} \quad (25)$$

$$\begin{aligned} o_{ijp}^y &\leq a_{ip}, a_{jp}, 1 - \lambda_{ijp}^y, 1 - \lambda_{jip}^y, \\ o_{ijp}^y &\geq a_{ip} + a_{jp} - 1 - \lambda_{ijp}^y - \lambda_{jip}^y, \end{aligned} \quad (26)$$

$$o_{ijp}^{xy} \leq o_{ijp}^x, \quad o_{ijp}^{xy} \leq o_{ijp}^y, \quad o_{ijp}^{xy} \geq o_{ijp}^x + o_{ijp}^y - 1. \quad (27)$$

Comma-separated upper bounds in Eqs. (25)–(26) denote one inequality for each listed right-hand side.

7 Support modes

Support is selected by `support.mode`. In `off` mode, none of the variables or constraints in this section are created. Otherwise the floor indicator is defined by

$$f_{jp} \leq a_{jp}, \quad z_{jp} \leq H(1 - f_{jp}), \quad z_{jp} \geq a_{jp} - f_{jp}. \quad (28)$$

Thus, for an assigned box, $f_{jp} = 1$ exactly at $z_{jp} = 0$.

A possible direct-support certificate satisfies

$$c_{ijp} \leq a_{ip}, a_{jp}, \lambda_{ijp}^z, o_{\min\{i,j\}, \max\{i,j\}, p}^{xy}, \quad (29)$$

$$z_{jp} - \bar{z}_{ip} \leq H(1 - c_{ijp}), \quad \bar{z}_{ip} - z_{jp} \leq H(1 - c_{ijp}). \quad (30)$$

Consequently, $c_{ijp} = 1$ implies same-pallet assignment, positive footprint overlap, i below j , and exact vertical contact $\bar{z}_{ip} = z_{jp}$. The converse need not be forced, but the support requirement forces enough valid certificates to cover every raised box.

7.1 Direct support

In `direct` mode, any positive horizontal overlap with at least one directly touching box is sufficient:

$$f_{jp} + \sum_{i \in \mathcal{I} \setminus \{j\}} c_{ijp} \geq a_{jp} \quad \forall j, p. \quad (31)$$

7.2 Fractional and full support

In **fraction** and **full** modes, the exact rectangle-intersection lengths are

$$q_{ijp}^x = \max\{0, \min(\bar{x}_{ip}, \bar{x}_{jp}) - \max(x_{ip}, x_{jp})\}, \quad (32)$$

$$q_{ijp}^y = \max\{0, \min(\bar{y}_{ip}, \bar{y}_{jp}) - \max(y_{ip}, y_{jp})\}, \quad (33)$$

$$A_{ijp} = q_{ijp}^x q_{ijp}^y. \quad (34)$$

The minima and maxima are inserted as Gurobi general constraints. The product in Eq. (34) is linearized exactly. In the default **compact** formulation, one overlap length is represented in binary, $q^x = \sum_k 2^k \beta_k$, each product $\beta_k q^y$ receives the usual three linear inequalities, and $A = \sum_k 2^k (\beta_k q^y)$. Configuration alternatives **lookup_1d** and **lookup_2d** enumerate feasible orientation and relative-offset states but produce the same exact overlap area.

The area counted as support is the contact-gated value

$$\begin{aligned} 0 &\leq S_{ijp} \leq A_{\min\{i,j\}, \max\{i,j\}, p}, \\ S_{ijp} &\leq A^{\max} c_{ijp}, \\ S_{ijp} &\geq A_{\min\{i,j\}, \max\{i,j\}, p} - A^{\max} (1 - c_{ijp}). \end{aligned} \quad (35)$$

Let $\hat{\alpha} = 1$ in **full** mode and let $\hat{\alpha} = \alpha = r/s$ in **fraction** mode. The implemented row is

$$s \sum_{i \neq j} S_{ijp} + r \bar{A}_j f_{jp} \geq r b_{jp} \quad \forall j, p. \quad (36)$$

For a raised assigned box, $f_{jp} = 0$, and this is exactly the required supported-area fraction. For a floor box, the term $r \bar{A}_j f_{jp}$ makes the row automatically feasible because $b_{jp} \leq \bar{A}_j$.

8 Symmetry-breaking constraints

All candidate pallets are interchangeable. The following always enforces a prefix of used pallet labels:

$$u_p \geq u_{p+1} \quad p = 0, \dots, P^{\max} - 2. \quad (37)$$

Three additional families are controlled by configuration (all are enabled in the supplied default file):

$$\sum_i v_i a_{ip} \geq \sum_i v_i a_{i, p+1} \quad \text{if } \text{order_pallet_loads}, \quad (38)$$

$$a_{0,0} = 1 \quad \text{if } \text{fix_first_item}. \quad (39)$$

For **order_identical_items**, boxes are grouped by the exact tuple

$$(\text{SKU}, (\ell_i, w_i, h_i), \text{weight}, \text{family}, \text{food}, \text{chemical}, \text{fragile}, \text{uprightOnly}, \text{priority}). \quad (40)$$

For consecutive indices (i, j) in any group,

$$\sum_{p \in \mathcal{P}} p a_{ip} \leq \sum_{p \in \mathcal{P}} p a_{jp}. \quad (41)$$

Define the injective integer encoding

$$\phi(x, y, z) := x(W + 1)(H + 1) + y(H + 1) + z. \quad (42)$$

On any pallet to which both boxes are assigned, their local positions are ordered:

$$\phi(x_{ip}, y_{ip}, z_{ip}) \leq \phi(x_{jp}, y_{jp}, z_{jp}) + M^{\text{loc}}(2 - a_{ip} - a_{jp}). \quad (43)$$

These constraints remove permutations of records treated as identical; they do not add a physical packing rule.

9 The reduced_exact variant

If `model_variant` is `reduced_exact`, identical candidate pallets no longer receive copies of a box's geometry. The assignment variables a_{ip} and pallet-use variables u_p remain, but orientation, coordinates, endpoints, direction relations, overlap lengths, areas, floor indicators, and contact variables drop their pallet index. Thus, for example,

$$r_{io} \in \mathbb{B}, \quad (x_i, y_i, z_i) \in \mathbb{Z}_{\geq 0}^3, \quad \lambda_{ij}^a \in \mathbb{B}, \quad c_{ij} \in \mathbb{B}. \quad (44)$$

Only orientations that individually fit within (L, W, H) are retained. The global containment equations are

$$\sum_o r_{io} = 1, \quad (45)$$

$$\bar{x}_i = x_i + \sum_o d_{io}^x r_{io} \leq L, \quad \text{quad} \bar{y}_i = y_i + \sum_o d_{io}^y r_{io} \leq W, \quad \text{quad} \bar{z}_i = z_i + \sum_o d_{io}^z r_{io} \leq H. \quad (46)$$

Non-overlap is activated only when a pair shares a candidate pallet:

$$\lambda_{ij}^x + \lambda_{ji}^x + \lambda_{ij}^y + \lambda_{ji}^y + \lambda_{ij}^z + \lambda_{ji}^z \geq a_{ip} + a_{jp} - 1 \quad \forall (i, j) \in \mathcal{E}, p. \quad (47)$$

The global direction variables are exact because their forward and reverse rows no longer need assignment deactivation terms.

For support, f_j indicates $z_j = 0$. A contact c_{ij} is permitted only if i and j have the same assignment vector; this is enforced for every p by

$$c_{ij} \leq 1 - a_{ip} + a_{jp}, \quad c_{ij} \leq 1 + a_{ip} - a_{jp}. \quad (48)$$

It also enforces $\bar{z}_i = z_j$. In direct mode it is gated by positive horizontal overlap. In fractional/full mode its gated intersection area is used exactly as above. The support row becomes

$$s \sum_{i \neq j} S_{ij} + r \bar{A}_j f_j \geq r \sum_{o \in \mathcal{O}_j} d_{jo}^x d_{jo}^y r_{jo}. \quad (49)$$

The objective, volume bounds, assignment, activation, and symmetry logic are otherwise the same. This variant is exact for the same discretized problem: because every pallet has identical dimensions, a box needs only one local coordinate and orientation for the single pallet to which it is assigned.

10 What is and is not guaranteed

The selected model is exact with respect to integer grid coordinates, its allowed orientation list, its configured support mode, and at most $P^{\max}$ candidate pallets. The conservative rounding may exclude a continuous packing that relies on sub-grid dimensions or positions. The volume bound strengthens the model but never replaces geometric non-overlap.

The model does not currently enforce pallet payload weight, center of mass, density, food-chemical separation, fragile-box restrictions, family compatibility, or retrieval accessibility. These fields remain available for future extensions and for reporting, but no such constraint should be inferred from the present solver output.